

Ramon McDargh-Mitchell

Full-Stack Software Engineer & ML Developer

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Education

University of California San Diego (UCSD)

Expected June 2027

B.S. Cognitive Science, Machine Learning & AI Specialization, Minor: Computer Science

GPA: 3.6

Relevant Coursework: Data Structures, Algorithms, Machine Learning, Neural Networks, Statistical Analysis

Technical Skills

Core Technologies: Next.js 16, React 19, TypeScript, Python, C++, Node.js

Backend & Databases: MySQL, Prisma ORM, Sanity CMS, REST APIs, GraphQL, Better Auth

Cloud & DevOps: Vercel, Oracle Cloud, AWS S3, Upstash Redis, Sentry, Rate Limiting

ML & AI: XGBoost, Random Forest, YOLOv8, OpenCV, OpenAI Whisper, SVM, Neural Networks, Scikit-learn, Pandas, NumPy, Matplotlib

Game Dev & VR: Unreal Engine 5, Virtual Reality Development, HCI, Simulation

Tools & Integrations: Stripe, Resend, Printful API, Zod, Git, Docker, CI/CD, Webhook Integration, GSAP

Professional Experience

Qualcomm Institute at UCSD | *VR Research Intern*

Fall 2025 - Present

- Developing AI-powered virtual reality avatars using Unreal Engine 5 for educational classrooms and museums
- Contributing AI object detection to VR SmartClassroom and NanoVerseVR for guided tutorials and nanoscale exploration
- Collaborating with faculty leaders on immersive technology research blending game development, ML, and HCI

Clutch Studio | *Founder & Full-Stack Developer*

Spring 2025 - Present

- Founded web development company serving real clients with full-stack solutions using Next.js 16, React 19, TypeScript
- Built comprehensive business platforms with admin dashboards, project management, payment processing, real-time data sync
- Deployed production applications with cloud infrastructure, monitoring, and CI/CD pipelines gaining me 3+ clients

In-N-Out Burger | *Food Service*

Spring 2021 - Fall 2025

- Prepared, cooked, and serviced food for customers with exceptional teamwork and customer service

Key Projects

Ascend - Hackathon & Ideathon Platform

Winter 2025 | [Visit Site](#)

- Built full-featured competition platform for organizations and companies using Next.js 16, React 19, Prisma ORM, AWS S3
- Implemented end-to-end workflow: organization management, event creation, team formation, judging, and live leaderboards
- Helping 3+ organizations save 10+ hours each by eliminating need to build custom infrastructure for competitions

Digital Revolution - Nonprofit E-commerce Platform

Summer 2025 | [Visit Site](#)

- Built full-stack nonprofit platform using Next.js 15 to expand STEM education access to underprivileged communities
- Implemented e-commerce with Stripe payments, Printful API fulfillment, NextAuth.js v5 authentication, donation system

ML Research: Empirical Comparison of AI Models

Fall 2024 | [View Project](#)

- Conducted comprehensive empirical study comparing XGBoost, Random Forest, and Neural Networks following NeurIPS format
- Built ML pipeline evaluating 3 classification and 2 regression datasets across multiple train-test splits with 10-fold CV
- Published research paper showing XGBoost outperformed on 2/3 classification and 1/2 regression datasets

Custom YOLOv8 Object Detection Model

Spring 2025 | [View Project](#)

- Trained domain-specific YOLOv8 model achieving 79.4% fitness score on 351 hand-labeled images
- Built complete ML pipeline: video recording, frame extraction, annotation, training, and deployment to detect personal objects

Technical Achievements & Leadership

- **VR Research:** Integrating machine learning into immersive educational experiences at prestigious research institute
- **Nonprofit Technology:** Built comprehensive e-commerce platform supporting STEM education with 50% profit allocation
- **Payment Integration:** Implemented complex payment flows including subscriptions, webhooks, and order management
- **UCSD Table Tennis:** Key player securing 1st place for JV team, defeating UCLA
- **HSF Scholar:** Was awarded the prestigious HSF Scholar award for my academic excellence
- **Upperclassmen Excellence in Mathematics:** Selected by the mathematics faculty at El Dorado High School during my senior year for my passion and performance in math courses.